

Chitin synthesis and degradation as targets for pesticide action.



Various pesticides are being used to destabilize, perturb, or inhibit crucial biochemical and physiological targets related to metabolism, growth, development, nervous communication, or behavior in pestiferous organisms. Chitin is an eukaryotic extracellular aminosugar biopolymer, massively produced by most fungal systems and by invertebrates, notably arthropods. Being an integral supportive component in fungal cell wall, insect cuticle, and nematode egg shell, chitin has been considered as a selective target for pesticide action. Throughout the elaborate processes of chitin formation and deposition, only the polymerization events associated with the cell membrane compartment are so far available for chemical interference. Currently, the actinomycetes-derived nucleoside peptide fungicides such as the polyoxins and the insecticidal benzoylaryl ureas have reached commercial pesticide status. The polyoxins and other structurally-related antibiotics like nikkomycins are strong competitive inhibitors of the polymerizing enzyme chitin synthase. The exact biochemical lesion inflicted by the benzoylaryl ureas is still elusive, but a post-polymerization event, such as translocation of chitin chains across the cell membrane, is suggested. Hydrolytic degradation of the chitin polymer is essential for hyphal growth, branching, and septum formation in fungal systems as well as for the normal molting of arthropods. Recently, insect chitinase activity was strongly and specifically suppressed by allosamidin, an actinomycetes-derived metabolite. In part, the defense mechanism in plants against invasion of pathogens is associated with induced chitinases. Chitin, chitosan, and their oligomers are able to act as elicitors which induce enhanced levels of chitinases in various plants. Lectins which bind to N-acetyl-D-glucosamine strongly interfere with fungal and insect chitin synthases. Plant lectins with similar properties may be involved in plant-pathogen interaction inter alia by suppressing fungal invasion.